STA 445 S26 Assignment 5

2026-03-17

```r
library(tidyverse)
library(lubridate)
library(mosaicData)
```

Problem 1

Convert the following to date or date/time objects.

a. September 13, 2010.

```r
make_date( 2010, 9, 13)
```

```
## [1] "2010-09-13"
```

b. Sept 13, 2010.

c. Sep 13, 2010.

d. S 13, 2010. Comment on the month abbreviation needs.

```r
mdy("Sept 13, 2010", "Sep 13, 2010", "S 13, 2010")
```

```
## [1] "2010-09-13" "2010-09-13" "2010-09-13"
```

Abbreviation is need as there are many different months that share the same first letter.

e. 07-Dec-1941.

f. 1-5-1998. Comment on why you might be wrong. Here i might be wrong because in the rest of the world the first number is the day and then the second is the month. from this format i have no way of knowing which data is correct.

g. 21-5-1998. Comment on why you know you are correct. There is no 21 month so this clearly is in day, month , year format.

```r
dmy("07-Dec-1941", "1-5-1998", "21-5-1998")
```

```
## [1] "1941-12-07" "1998-05-01" "1998-05-21"
```

h. 2020-May-5 10:30 am

```r
ymd_hm("2020-May-5 10:30 am")
```

```
## [1] "2020-05-05 10:30:00 UTC"
```

i. 2020-May-5 10:30 am PDT (ex Seattle)

```r
ymd_hm("2020-May-5 10:30 am",tz = "America/Los_Angeles")
```

```
## [1] "2020-05-05 10:30:00 PDT"
```

j. 2020-May-5 10:30 am AST (ex Puerto Rico)

```r
ymd_hm("2020-May-5 10:30 am",tz = "America/Puerto_rico")
```

```
## [1] "2020-05-05 10:30:00 AST"
```

Problem 2

Using just your date of birth (ex Sep 7, 1998) and today's date calculate the following:

a. Calculate the date of your 64th birthday.

```r
birthday <- mdy("April, 29, 2005")

birthday + dyears(64)
```

```
## [1] "2069-04-29"
```

b. Calculate your current age (in years).

```r
current.age = interval(birthday,  lubridate::today() )

as.period(current.age) %>%
  year()
```

```
## [1] 20
```

c. Using your result in part (b), calculate the date of your next birthday.

```r
time_till_nextBirthday<- interval(current.age, dyears(21)) %>%
  as.period()

time_till_nextBirthday
```

```
## [1] "1m 13d 6H 0M 0S"
```

d. The number of *days* until your next birthday.

```r
days.till <- interval(current.age, dyears(21)) %>%
  as.period(unit = "days") %>%
  day()


days.till
```

```
## [1] 43
```

e. The number of *months* and *days* until your next birthday.

```r
MD.till <- interval(current.age, dyears(21)) %>%
  as.period() %>%
  month()


MD.till
```

```
## [1] 1
```

Problem 3

Suppose you have arranged for a phone call to be at 3 pm on May 8, 2015 at Arizona time. However, the recipient will be in Auckland, NZ. What time will it be there?

```r
AZ <- mdy_hm( "May 8, 2015, 15:00" , tz = "US/Arizona")
AK <- with_tz(AZ,"Pacific/Auckland")

AK
```

```
## [1] "2015-05-09 10:00:00 NZST"
```

Problem 4

It turns out there is some interesting periodicity regarding the number of births on particular days of the year.

a. Using the `mosaicData` package, load the data set `Births78` which records the number of children born on each day in the United States in 1978. Because this problem is intended to show how to calculate the information using the `date`, remove all the columns *except* `date` and `births`.

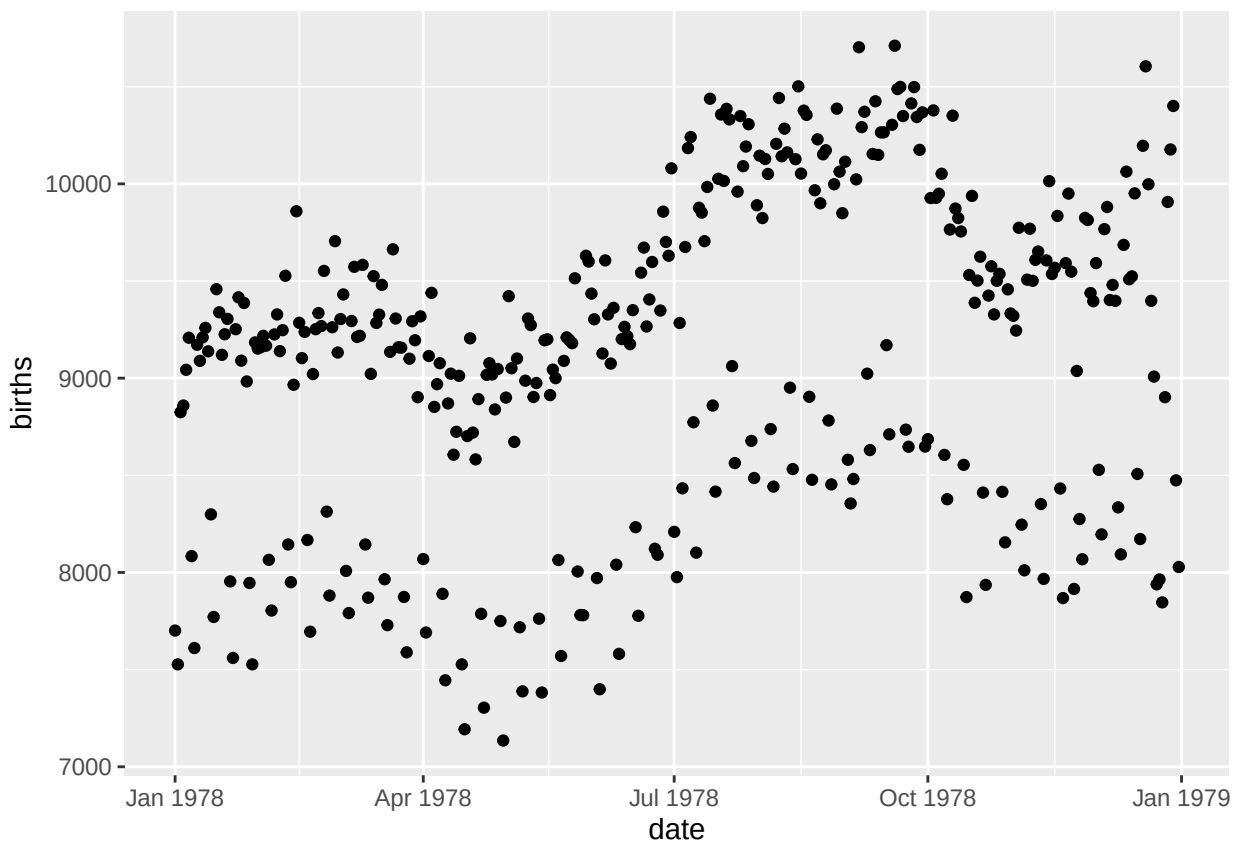
```r
data("Births78")
```

```r
my.78 <- Births78

my.78 <- my.78  %>%
  select("date", 'births')
```

b. Graph the number of `births` vs the `date` with date on the x-axis. What stands out to you? Why do you think we have this trend?

```
ggplot(my.78,
       aes(x = date,
           y = births))+
  geom_point()
```



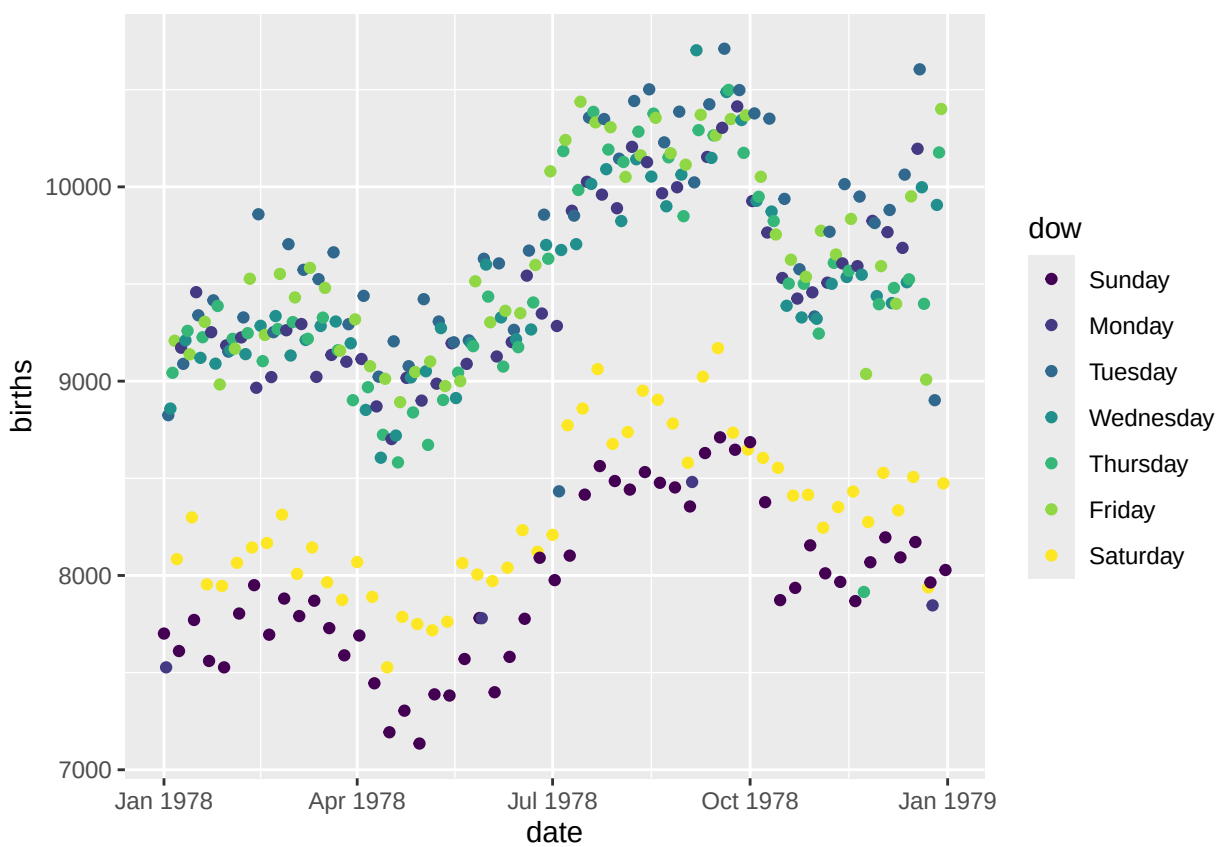
c. To test your assumption, we need to figure out the what day of the week each observation is. Use `dplyr::mutate` to add a new column named `dow` that is the day of the week (Monday, Tuesday, etc). This calculation will involve some function in the `lubridate` package and the `date` column.

```
my.78 <- rowwise(my.78) %>%
  mutate( dow = wday(date,label=TRUE, abbr=FALSE))

my.78
```

```
## # A tibble: 365 x 3
## # Rowwise:
##    date       births dow
##    <date>      <int> <ord>
##  1 1978-01-01   7701 Sunday
##  2 1978-01-02   7527 Monday
```

```
##   3 1978-01-03   8825 Tuesday
##   4 1978-01-04   8859 Wednesday
##   5 1978-01-05   9043 Thursday
##   6 1978-01-06   9208 Friday
##   7 1978-01-07   8084 Saturday
##   8 1978-01-08   7611 Sunday
##   9 1978-01-09   9172 Monday
##  10 1978-01-10   9089 Tuesday
## # i 355 more rows
```

d. Plot the data with the point color being determined by the day of the week variable.

```
ggplot(my.78,
       aes(x = date,
           y = births,
           color = dow))+
  geom_point()
```